

MURAL-Fi: Multi-User Respiration Authentication Leveraging WiFi

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Integrating WiFi communication with vital signs sensing, particularly under the emerging 802.11bf standard, offers a promising pathway toward next-generation authentication systems, providing non-intrusive, continuous, and scalable security. However, current WiFi-based respiration authentication approaches are limited to single-user scenarios, whereas existing multi-user respiration monitoring technologies either lack identity awareness or require users to remain completely stationary. In this paper, we present MURAL-Fi, the first system to achieve multi-user respiration authentication using WiFi, addressing three critical requirements: multi-user support, identity-aware capability, and involuntary-motion tolerance. Unlike conventional methods that monitor the movement of the highest-energy point within the respiration frequency band of the WiFi spectrum, MURAL-Fi introduces a novel baseline region variation exploration technique. This technique identifies a spatial region of points corresponding to the chest cavities and extracts respiration signals by analyzing strength variations within these regions. Specifically, we iteratively employ baseline region identification coupled with interference suppression utilizing spatial spectral information derived from WiFi signals. This iterative refinement progressively enhances the precision of chest-cavity localization and robustly mitigates various interferences, enabling identity-aware respiration pattern extraction despite involuntary body movements. Subsequently, we perform multi-level feature analysis to derive user-specific respiratory characteristics and utilize a triplet loss-based Siamese Neural Network to authenticate users. Through extensive experiments with 24 participants across diverse environments, we demonstrate that MURAL-Fi achieves high authentication accuracy, maintaining superior performance across challenging scenarios, including resistance to mimic attacks.

CCS Concepts: • **Security and privacy** → **Authentication**.

Additional Key Words and Phrases: Wireless sensing, Authentication, Multi-User

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1 INTRODUCTION

Integrating WiFi communication with vital signs sensing, such as respiration, presents a promising approach for advancing next-generation authentication systems, especially with the emerging 802.11bf standard that enhances WiFi sensing capabilities [32]. By leveraging the ubiquity and scalability of WiFi, this method enables

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Table 1. WiFi-based Respiration Sensing Systems.

Technology	Spatial-Information Usage	Multi-User Support	Identity-Aware Capacity	Involuntary-Motion Tolerance
Cuvrb [25]	×	×	×	×
Multisense [48]	×	✓	×	×
PhaseBeat [43]	×	✓	×	×
BreathTrack [49]	✓	✓	×	×
TR-Breath [8]	✓	✓	×	×
Wiphone [27]	×	✓	✓	×
MUSE-Fi [18]	×	✓	✓	×
SpaceBeat [23]	✓	✓	✓	×
MURAL-Fi (Ours)	✓	✓	✓	✓

non-intrusive, user-friendly, and continuous respiration-based authentication in multi-user scenarios. Unlike traditional methods that rely on passwords, tokens, or activity recognition [4, 21], respiration-based authentication eliminates the need to memorize credentials [30], carry external devices [7, 15, 19, 47], or perform continuous actions [21, 31], allowing users to focus on other tasks while maintaining seamless and uninterrupted security [2]. Furthermore, WiFi-based systems complement camera- and radar-based solutions [50] by providing ubiquitous authentication due to their widespread deployment and their capability to function effectively in non-line-of-sight (NLoS) conditions.

However, most existing WiFi-based respiration authentication systems are limited to single-user scenarios and cannot support multiple users simultaneously [1, 7, 25]. An intuitive approach is to extend recent multi-user respiration monitoring technologies [8, 43, 48, 49], as listed in Table 1, for authentication. While these technologies can distinguish respiration patterns from multiple individuals, they fail to associate each pattern with its corresponding user. This limitation prevents these monitoring systems from achieving the identity awareness essential for authentication in multi-user environments.

Even with the latest advancements in respiration monitoring technologies, systems like MUSE-Fi [18], Wiphone [27] and SpaceBeat [23] have introduced identity-aware capabilities, yet their practicality for real-world authentication remains limited. To maximize the distinction between respiration patterns and environmental interference, these systems require individuals to remain completely stationary during the monitoring process. While users can generally cooperate with authentication protocols by minimizing large movements, eliminating involuntary motions entirely, such as leg shaking, finger tapping, or speech, is often impossible, as these motions often stem from physiological conditions, psychological states (e.g., restlessness or anxiety), or inherent behavioral traits. Unlike environmental interference, these motions occur near the chest cavity and significantly disrupt respiration pattern extraction [5, 6]. Consequently, the strict requirement for users to remain still limits the practicality of these advanced systems in real-world authentication scenarios.

In this paper, we propose MURAL-Fi, the first system to achieve multi-user respiration-based authentication using WiFi. Unlike existing respiration monitoring systems, MURAL-Fi is specifically designed to fulfill three critical requirements for authentication: *multi-user support*, *identity-aware capability*, and *involuntary-motion tolerance*. By leveraging spatial information and addressing these requirements [50], MURAL-Fi delivers robust and accurate authentication, ensuring practical applicability in complex conditions.

The challenge of MURAL-Fi lies in accurately determining chest movements. Traditional radar-like methods rely on the assumption that chest movements follow a periodic back-and-forth motion, which aligns with the frequency range of human respiration. These approaches identify and track the highest energy peak in the WiFi spectrum within this range, presuming that the trajectory of this spectral peak accurately reflects the

cyclic motion of the chest. [8, 23, 49]. However, our research reveals that in environments with multiple users, involuntary movements, and various forms of interference, the highest energy peak on the WiFi spectrum does not always correspond to the actual chest cavity location. Such interference can cause spurious high-energy points in the spectrum, leading to misleading peak trajectories. Therefore, these trajectories often deviate from the assumed periodic motion and instead appear random and discontinuous.

To tackle these challenges, MURAL-Fi leverages modern WiFi MIMO capabilities to construct fine-grained spatial information, enabling the identification of user-specific regions in the WiFi spectrum. Its core innovation lies in a statistical identity-aware approach that robustly characterizes these regions by aggregating several strong energy points, rather than tracking a single trajectory. This region is adaptively associated with a specific user, offering a statistically robust and personalized representation of the chest cavity. Once the approximate chest cavity region is identified, MURAL-Fi extracts the user's dominant respiratory component by analyzing energy variations within the region. This approach is chosen because WiFi reflections are highly susceptible to environmental factors, which can cause the entire region to shift. Despite these positional shifts, the temporal variations within the region provide consistent and reliable indicators of respiratory patterns.

To achieve a more precise identification of the chest cavity region, MURAL-Fi incorporates an iterative region refinement method. Instead of simply defining an initial region, this process dynamically adjusts the region based on the spatial distribution of the respiratory component's intensity, progressively enhancing the accuracy of the target area. Concurrently, residual environmental interference within the region is suppressed as part of the refinement process. This ensures that the region is continuously optimized while minimizing the impact of misleading energy points. Through iterative refinement and interference mitigation, MURAL-Fi enhances its ability to extract accurate user-specific respiration patterns.

After extracting the user-specific respiration patterns [7, 10, 17], we perform multi-level feature analysis. These multi-level respiration-related features are then structured into an image-like input and fed into a triplet loss-based Siamese Neural Network (SNN) [34] for authentication. Unlike traditional deep learning models [7], a siamese network can achieve effective performance with only a few instances per class. By leveraging triplet loss, the network learns a more robust and discriminative embedding space through relative similarity comparisons compared to twin siamese architectures. [16]. The network minimizes embedding distances between samples from the same user while maximizing those from different users, ensuring that the extracted features are dominated by user-specific respiratory signals with minimal noise. This design makes the authentication process accurate and robust to environmental variations.

We conducted extensive experiments involving 24 participants across a variety of environments, including laboratories, classrooms, and conference rooms. The evaluation encompassed diverse scenarios, such as multi-user settings, mobile interference, varying distances, and differing orientations. The experimental results demonstrate that MURAL-Fi delivers high robustness and accuracy in authentication tasks. The key contributions of this work are summarized as follows:

- We propose MURAL-Fi, the first system to achieve multi-user respiration-based authentication leveraging WiFi. MURAL-Fi addresses the critical requirements of multi-user support, identity-aware capability, and involuntary motion tolerance, ensuring practical applicability in complex conditions.
- We introduce a novel statistical approach for robustly identifying the chest cavity region by evaluating energy variations within a defined region instead of focusing on the highest energy peak. This is complemented by an iterative region refinement method that progressively enhances accuracy while mitigating environmental interference, enabling reliable respiration pattern extraction.
- We validate MURAL-Fi through extensive experiments involving 24 participants across diverse environments. Our evaluation demonstrates its high robustness and accuracy across diverse scenarios.

2 MODELS AND ASSUMPTIONS

In this section, we first introduce the threat model and assumptions. Then, we discuss our system model.

2.1 Threat Model and Assumptions

Our threat model considers adversaries whose goal is to deceive the respiration WiFi sensing-based authentication system. We assume that adversaries can position themselves within the effective WiFi sensing coverage area and deliberately manipulate reflected signals to gain unauthorized access. Based on their level of prior knowledge and attack strategy, we categorize threats into two primary types:

Brute-Respire Attack: In this scenario, the adversary has no specific knowledge about the respiration patterns of legitimate users. The attacker enters the sensing region and brute-forces arbitrary breathing patterns, hoping these patterns may coincidentally resemble those of an enrolled user and bypass authentication.

Mimic Attack: In this scenario, the adversary is assumed to have observed the legitimate user and obtained detailed knowledge of their respiratory behaviors—including rate, amplitude, posture, and involuntary motions. The attacker seeks to replicate respiration patterns to deceive the system. We further assume that adversaries may possess a similar body shape to the legitimate user and have knowledge of the authentication mechanism itself, including its reliance on WiFi-based sensing and signal variation analysis. This enables them to introduce controlled interference and manipulate signal reflections to dominate the sensing process with forged patterns that closely resemble those of the legitimate user, thereby deceiving the system into granting unauthorized access.

2.2 System Model

Our proposed system, MURAL-Fi, utilizes WiFi devices to achieve robust authentication by leveraging the unique respiration patterns of users. The **enrollment process** operates in a controlled, interference-free indoor environment to establish a reliable respiration profile for each user. The user sits one meter from the WiFi receiver, facing the device in a static posture. WiFi transceivers capture reflected signals from multiple angles to extract high-fidelity respiration features. Each enrollment session lasts about 300 seconds, during which clean and stable data are collected to ensure precise feature extraction. The acquired respiration-related features are then processed into a user-specific representation and stored into profile. Enrollment is required only once to build their profile and does not need to be repeated in every new environment. Once this profiling is completed, the system transitions to **authentication process** used in a multi-user environment. Here, WiFi signals propagate within the space, and the receiver distinguishes the target user from others based on their unique respiration pattern. Upon successful identification, the system grants access to the authorized user. The setup involves deploying a WiFi transmitter and receiver, where the receiver captures the variations in WiFi reflections caused by respiration. This WiFi-based deployment ensures the system is scalable and easy to integrate into existing environments.

3 PRELIMINARIES OF WIFI SENSING

Human respiration manifests as subtle periodic movements of the chest cavity, producing weak but detectable perturbations in wireless signals. As illustrated in Fig. 1a, as the chest moves during respiration, it causes slight displacements in the WiFi signal reflection path, leading to small but measurable changes in the path's length and angle. To accurately capture these minute variations, we leverage the rich multidimensional features contained within WiFi Channel State Information (CSI). CSI characterizes how WiFi signals propagate and interact with the environment, providing diverse features that can be analyzed to isolate and authenticate respiratory patterns.

As illustrated in Fig. 1b, the key channel features of CSI utilized for respiration-based authentication are:

Angle of Arrival (AoA): AoA characterizes the direction of incoming WiFi signals at the receiver's antenna array [46]. In our three-dimensional sensing system, AoA is described by two components: the azimuth angle

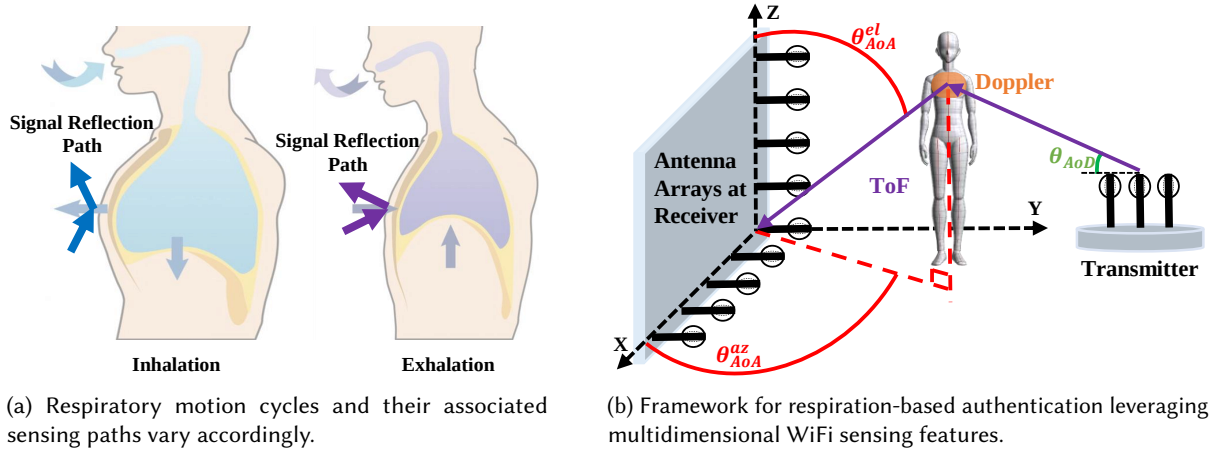


Fig. 1. Illustration of the rationale behind the respiration-based authentication system leveraging WiFi signals.

(θ_{AoA}^{az}) in the horizontal plane and the elevation angle (θ_{AoA}^{el}) in the vertical plane. The phase shift (Φ_{AoA}) observed across the antenna array elements depends on these angles and can be computed as:

$$\Phi_{AoA}^{az} = e^{-j2\pi d \cos(\theta_{AoA}^{az}) \sin(\theta_{AoA}^{el}) / \lambda} \quad (1)$$

$$\Phi_{AoA}^{el} = e^{-j2\pi d \cos(\theta_{AoA}^{el}) / \lambda} \quad (2)$$

where λ denotes the WiFi signal wavelength, and d equals the distance between adjacent antennas (typically $\frac{\lambda}{2}$). The phase shifts Φ_{AoA}^{az} and Φ_{AoA}^{el} capture the signal's spatial propagation along the azimuth and elevation directions, respectively.

Angle of Departure (AoD): AoD characterizes the direction of outgoing WiFi signals from the transmitter's antenna array [36]. Similar to AoA, AoD is described by two components: the azimuth angle (θ_{AoD}^{az}) in the horizontal plane and the elevation angle (θ_{AoD}^{el}) in the vertical plane. Given the reciprocal nature of signal propagation, the phase shifts for AoD (Φ_{AoD}) follow the same mathematical formulation as AoA, with the corresponding AoD angles.

Time of Flight (ToF): ToF characterizes the propagation time of WiFi signals between the transmitter and receiver, enabling the estimation of path lengths to reflecting surfaces [22]. In a WiFi system, ToF can be derived from the phase shift (Φ_{ToF}) across subcarriers:

$$\Phi_{ToF} = e^{-j2\pi f_{\delta} \tau} \quad (3)$$

where f_{δ} denotes the frequency spacing of WiFi, and τ represents the signal propagation delay.

Doppler Velocity: This parameter characterizes the frequency shifts induced by motion, making it particularly suited for detecting the periodic movements associated with breathing. The Doppler phase shift (Φ_{Dop}) in the received signal can be computed as:

$$\Phi_{Dop} = e^{-j2\pi t_{\delta} v / \lambda} \quad (4)$$

where v represents the instantaneous velocity of chest movement, and t_{δ} denotes the time interval between consecutive measurements obtained using WiFi packets.

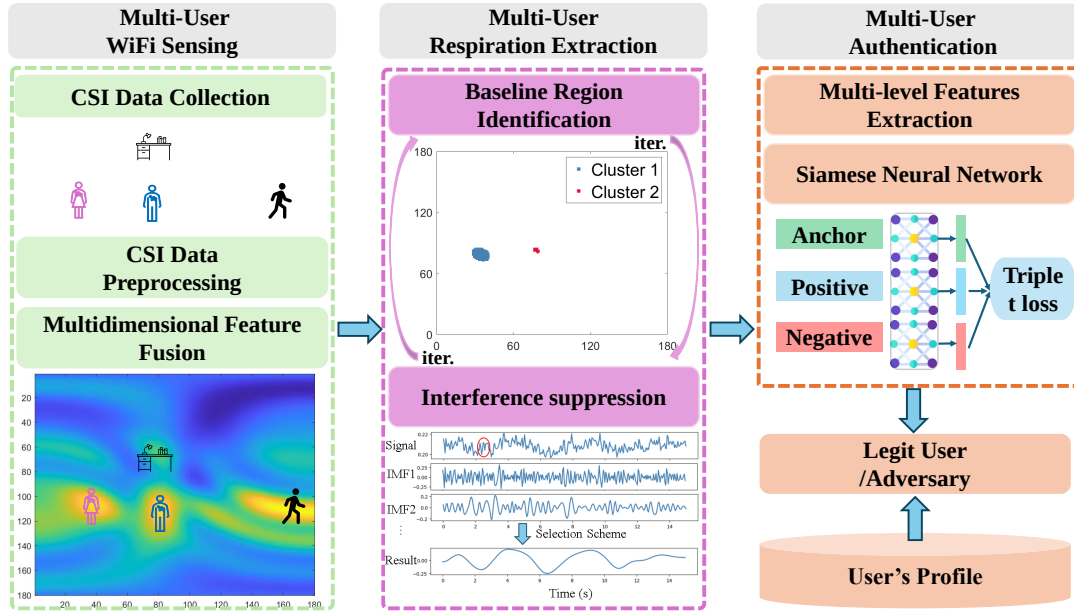


Fig. 2. System architecture of MURAL-Fi.

These multidimensional parameters work in concert to enable robust respiration detection and authentication. AoA and AoD provide precise spatial localization of individual users, allowing the system to distinguish between multiple breathing subjects. ToF helps isolate respiratory reflections from environmental multipath, while Doppler velocity analysis captures the speed information of users' breathing cycles, typically within the range of 0–1 m/s. This range preserves respiratory features while suppressing certain noise components. It can effectively suppress high-speed noise caused by body movements, environmental disturbances, or unrelated activities. This comprehensive approach ensures reliable extraction of respiratory patterns even in challenging multi-user scenarios.

4 SYSTEM OVERVIEW

This section outlines the system architecture in Fig. 2. It begins with WiFi CSI data collection, followed by data preprocessing to eliminate distortions, enabling accurate extraction of spatial information. Multidimensional feature fusion across AoA, AoD, ToF, and Doppler enhances the clarity and separability of respiration signals, even in multi-user and motion-interference scenarios. Rather than tracking the movement of the highest-energy point, MURAL-Fi identifies respiration-related baseline regions. An iterative refinement process further localizes the chest region by isolating respiratory energy and suppressing involuntary and cross-user interference. From the refined region, the system extracts multi-level respiration features. These features are fed into a triplet-loss-based SNN, which embeds them into discriminative feature spaces. By comparing the embedding distances with registered user profiles in the feature space, our system achieves accurate and robust multi-user authentication based on identity-aware respiration signals.

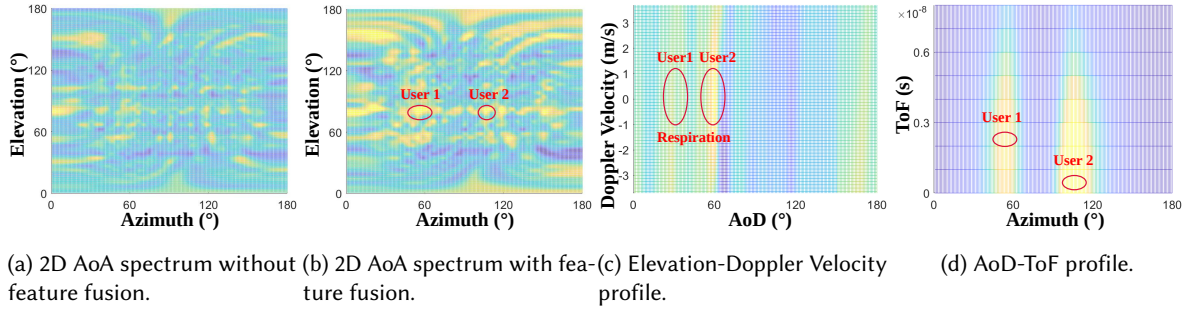


Fig. 3. Effect of multidimensional feature fusion on the AoA spectrum. The fusion process significantly enhances the visibility of respiration reflections from multiple users, enabling more accurate detection and differentiation. The AoD-Doppler Velocity and Azimuth-ToF profiles demonstrate that multidimensional features can effectively capture user-specific respiration patterns.

5 MULTIDIMENSIONAL FEATURE FUSION

Chest cavity movements due to respiration tend to produce weak and subtle reflections in WiFi signals, particularly when compared to stronger echoes from static objects or larger human movements. Relying on a single dimension, such as AoA or Doppler Velocity, can easily overlook these faint reflections or struggle to separate them from interference. For instance, as shown in Fig. 3c, Doppler alone might captures rhythmic breathing motion (Doppler velocity around 0–1 m/s) for both users, but without localization from AoA/AoD or distance information from ToF, one cannot pinpoint that motion as being tied to the user’s chest, as illustrated in Fig. 3d.

To address this, we fuse multiple dimensions (AoA, AoD, ToF, and Doppler Velocity) into a unified representation, providing more robust evidence of respiratory motion. We first preprocess the raw CSI data to remove phase inconsistencies caused by hardware imperfections, such as STO and PDD by applying an optimal linear fitting to the unwrapped CSI phase across subcarriers and antennas. The fitted linear component is then subtracted from each packet’s CSI phase, eliminating STO and PDD induced phase variation across packets for accurate AoA and ToF estimation [22]. After correction, we perform multidimensional feature fusion through the following steps: **Joint Parameter Estimation via MUSIC**. To operationalize this fusion, we first formulate a steering vector::

$$\mathbf{A}(\theta_{AoA}, \theta_{AoD}, \tau, v) = \text{vec}(\Phi_{AoA}^{N_a} \otimes \Phi_{AoD}^{N_d} \otimes \Phi_{ToF}^{N_t} \otimes \Phi_{Dop}^{N_v}) \quad (5)$$

where N_a , N_d , N_t , and N_v represent the size in AoA, AoD, ToF, and Doppler Velocity dimensions respectively, and $\otimes$ denotes the Kronecker product. The concatenation (via vec) creates a unified representation of the complete path signature. This formulation captures the interdependencies across all dimensions. In particular, the correlation of AoA/AoD with ToF can help localize reflections to a specific angular sector and range, while Doppler Velocity highlights minute rhythmic movements within that region.

Using this steering vector, we apply the MUSIC algorithm [33] to estimate the multidimensional parameters jointly:

$$P(\theta_{AoA}, \theta_{AoD}, \tau, v) = \frac{1}{\mathbf{A}^H \mathbf{E}_N \mathbf{E}_N^H \mathbf{A}} \quad (6)$$

where $\mathbf{E}_N$ represents the noise subspace. This joint estimation generates an multidimensional array $P \in \mathbb{R}^{N_a \times N_d \times N_t \times N_v}$, where each dimension corresponds to AoA, AoD, ToF, and Doppler Velocity respectively. Unlike separate estimations that must independently infer angles, distances, or velocity, this joint approach allows the

algorithm to leverage correlations across all dimensions. In doing so, it more effectively distinguishes between real chest-cavity motions and other random or periodic environmental signals.

Respiration Pattern Enhancement. After constructing the multidimensional power spectrum $P(\theta_{AoA}, \theta_{AoD}, \tau, v)$, our goal is to produce a spatial map that highlights respiratory reflections. To this end, we project the multidimensional power spectrum onto a 2D AoA plane through dimension reduction:

$$P_{enh}(\theta_{AoA}^{az}, \theta_{AoA}^{el}) = \sum_{\theta_{AoD}} \sum_{\tau} \sum_v P(\theta_{AoA}, \theta_{AoD}, \tau, v) \quad (7)$$

By summing across AoD, ToF, and Doppler Velocity, we retain the integrated energy that aligns with chest-cavity motion while mitigating unrelated noise. Intuitively, each point in the resulting 2D map represents the cumulative energy of signals arriving at a particular azimuth and elevation, aggregated over the other dimensions. Consequently, any faint but consistent respiration signal gains enhanced prominence, as illustrated in Fig. 3. The difference between non-fusion (Fig. 3a) and fusion-enhanced (Fig. 3b) AoA maps demonstrates how leveraging multiple dimensions intensifies the visibility of respiratory reflections from multiple users. In this section, the input is the raw CSI data, and the output is continuous 2D AoA spectrums. The size of the input CSI data depends on the number of transmit antennas, receive antennas, and subcarriers over time. The size of generated 2D AoA spectrum depends on azimuth and elevation resolutions over time. The CSI sampling rate is 1000 packets per second, while the generated spectrum has a frame rate of 50 frames per second. Each data segment covers a 30-second time window.

6 BASELINE REGION IDENTIFICATION

After significantly enhancing respiratory signals with multidimensional feature fusion, extracting stable respiration patterns remains challenging due to interference from static objects and other moving entities in the environment.

In our approach, rather than relying on tracking the peak trajectories, we establish a baseline region that encapsulates the core information of chest cavity movement. Due to the feature fusion, the baseline region effectively aggregates information related to AoD, ToF, and Doppler, making it possible to extract these parameters without needing to follow the detailed trajectory of the chest cavity. This approach allows us to observe the variations within the baseline region, which implicitly carries the necessary respiratory features. Although these parameters might overlap, each has its unique characteristics that can be extracted independently. By focusing on the baseline region, we simplify the process and enhance the robustness of our system, providing a more reliable way to capture respiratory patterns without being affected by the complexities of tracking movement trajectories.

We identify a consistent baseline region that represents the core area of chest cavity movement by using:

BNR Calculation: The first step in our approach is the calculation of a BNR (Breathing-to-Noise Ratio) across the entire spectrum [47]. This ratio is determined by comparing the energy at respiratory frequencies with background noise levels. We first isolate the frequency band corresponding to typical human respiration. For each point in this band, we measure the signal energy and compare it to the average energy of the surrounding noise. Instead of using a fixed threshold, we apply an adaptive thresholding mechanism based on the distribution of noise levels, ensuring that only points with a significantly higher BNR are selected. This adaptive threshold dynamically adapts to environmental variations.

Spatial Clustering: The second step involves employing a spatial clustering algorithm to define comprehensive baseline regions. We identify candidate points that exhibit a high BNR from the previous step. Using a density-based spatial clustering technique, we group these high BNR points into clusters. This technique is particularly effective as it does not require the number of clusters to be predefined and can identify regions based on the

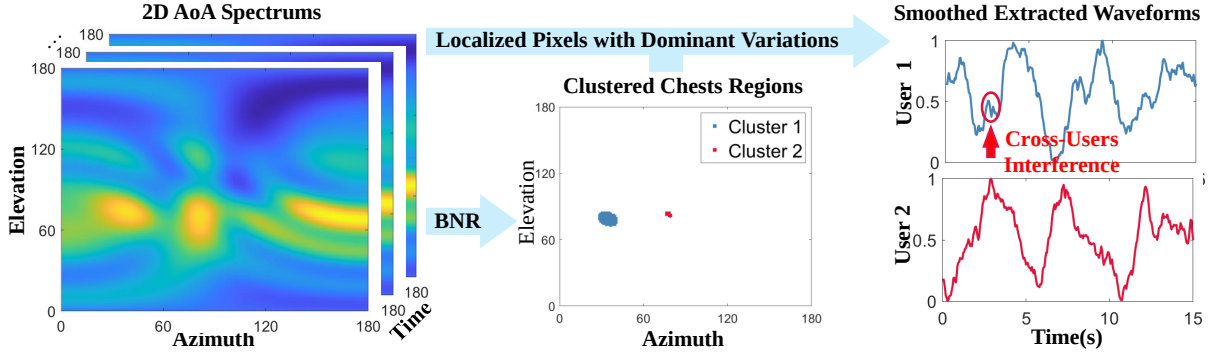


Fig. 4. Baseline region identification in spectrums. Instead of following high-energy peaks, MURAL-Fi isolates respiration by analyzing temporal variations within clustered chest regions. The red arrow indicates weak cross-user interference that must be suppressed to preserve respiration signal quality.

density of points. Clusters formed by the algorithm are verified to ensure they represent a consistent region of chest cavity movement. Points that do not form dense clusters are discarded as noise.

7 ITERATIVE REGION REFINEMENT AND RESIDUAL INTERFERENCE MITIGATION

Extracting respiration from continuous 2D AoA spectrums remains challenging even after identifying a baseline region. Two key challenges must still be addressed: first, refining the initial baseline region to precisely localize the chest cavity, ensuring it tightly corresponds to the individual's respiratory motion; second, mitigating residual interference, including environmental noise, cross-user interference, and involuntary motions, which can obscure or distort the true breathing patterns [51]. To address these challenges, MURAL-Fi employs an iterative refinement strategy that progressively hones the baseline region by dynamically adjusting its boundaries and suppressing misleading energy points.

7.1 Refinement of Chest Cavity Regions

Rather than fixing the chest cavity region at the outset, MURAL-Fi adopts an iterative region refinement process. Initially, a coarse region is established in the 2D AoA spectrums around each user's estimated location. This region is progressively narrowed down through recalculating the ratio between breathing energy and overall energy (BNR) [47]. Over successive iterations, pixels that fail to maintain the BNR threshold are removed, and the threshold itself is dynamically adjusted based on the evolving signal distribution. Convergence is reached when the region's size and shape stabilize, indicating that the remaining points reliably capture chest motion.

Figure 4 illustrates this process. User 1 and User 2 start with coarse regions influenced by cross-user interference and environmental noise. As the iterations proceed, points with dominant variations in breathing patterns emerge, forming distinct clustered chest regions. This clustering step ensures that interference is minimized, and each user's respiratory motion is captured independently.

7.2 Dynamic Decomposition for Interference Suppression

With the refined region in place, MURAL-Fi next mitigates residual interference to ensure that subsequent analyses focus on the true breathing signals. Respiratory signals are intrinsically non-stationary and can exhibit time-varying frequency and amplitude [39], making them poorly suited for simple bandpass filtering [18, 26]. Frequency-based bandpass filtering methods [18] fail to remove cross-user interference because other users'

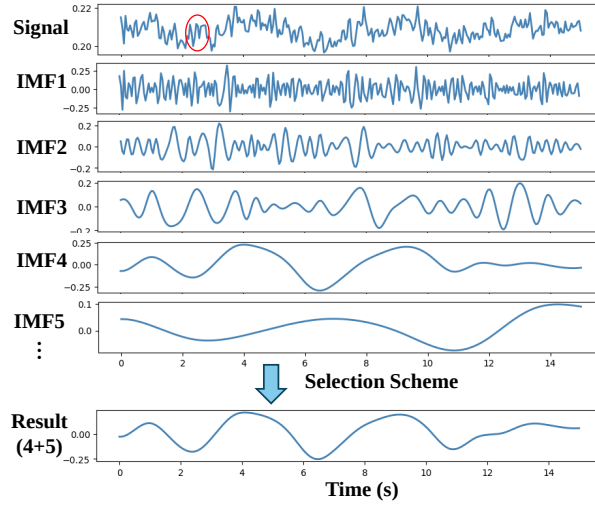


Fig. 5. Illustration of the proposed dynamic interference suppression method. By adaptively estimating the noise energy ratio, MURAL-Fi decomposes noise and cross-user interference (highlighted in the red circle) into non-respiratory IMFs, which are subsequently discarded.

breathing occupies the same frequency range as that of the target user breathing. As a result, we turn to an adaptive noise Empirical Mode Decomposition (EMD)–based approach that decomposes the measured signal into a set of intrinsic mode functions (IMFs) and adaptively excludes interfering components. As shown in fig. 5, cross-user interference, characterized by subtle oscillations, is decomposed into IMFs and often mixed with noise, such as in IMFs 2 and 3, making it not to be selected by our selection scheme. This demonstrates the effectiveness of our adaptive noise EMD method in isolating the true respiratory signal while suppressing environmental and cross-user interference.

Specifically, similar to Ensemble Empirical Mode Decomposition (EEMD) [38], we incorporate multiple noise realizations into the measured signal before performing EMD to avoid mode mixing. However, in contrast to standard EEMD, our added noise is *adaptive*: its strength is tuned based on the relative power of the interference and the respiratory component. This step yields a more robust separation of the IMFs. Formally, we generate multiple versions of the original signal x by adding noise vectors N_i , giving $x_i = x + N_i$. After decomposing each x_i into IMFs, we compute a unique first residue $r_1 = x - \text{IMF}_1$. Subsequent IMFs are then extracted by iteratively adding scaled noise terms:

$$\text{IMF}_2 = O_1(r_1 + \sigma_1 \cdot O_1(N)) \quad (8)$$

where σ_j is a dynamic scaling factor chosen to match the interference strength at each decomposition stage, and $O_j(\cdot)$ represents the operator that extracts the j -th IMF from the signal. Iterating this procedure prevents IMFs from blending respiratory information with environmental interference.

After obtaining the IMFs, MURAL-Fi systematically evaluates possible combinations of them to isolate the true respiration signal. We compute the BNR for each candidate reconstruction of the breathing waveform. The configuration with the highest BNR is selected as the final respiratory signal. By focusing on maximizing BNR across the combinations of IMFs, we ensure that residual noise or incomplete decompositions do not degrade the recovered breathing patterns.

Algorithm 1 Multi-User Respiration Extraction Iterative Framework**Input:** 2D AoA spectrums $P(\theta_{az}, \theta_{el}, timestamps)$ **Output:** Extracted respiration signals $S(timestamps)$ **1. Breathing-to-Noise Ratio Analysis****for** each point (i, j) in spectrums **do**

$$|B_s[i, j]| = \frac{\sum_{n=0}^{t-1} |P_{i,j}[n]e^{-j\frac{2\pi kn}{t}}|}{\sum_{n=0}^{t-1} |P_{i,j}[n]|},$$

 $|i = 0, 1, \dots, \theta_{az}, j = 0, 1, \dots, \theta_{el}, n = 0, 1, \dots, t - 1$ **end for****2. Region Formation**Retain points sets $L(i, j)$ whose B_s exceed the thresholdCluster retained points $L(i, j)$ to identify N individualsThe clustered $L(i, j)$ define the initial chest cavity regions**3. Iterative process****while** the number of individuals N and locations $L(i, j)$ of chest cavity regions changes **do**| Perform **dynamic decomposition** on $P(L(i, j), timestamps)$ | Update $P(L(i, j), timestamps)$ | Repeat **breathing-to-Noise ratio analysis**| Repeat **region formation****end while**Apply PCA on $P(L(i, j), timestamps)$ to reduce dimensionality and obtain $S(timestamps)$

7.3 Iterative Refinement for Stabilized Regions

Because environmental interference and movement can change over time, MURAL-Fi updates both the localized region and the interference-mitigated signals in an iterative feedback loop. As shown in Algorithm 1, after each round of interference mitigation, we re-compute the BNR within the 2D AoA spectrums and prune any newly emerged points of interference. Once the region stabilizes in both size and shape, we aggregate signals from the remaining pixels using Principal Component Analysis (PCA). PCA is ideal for extracting a single dominant motion pattern, respiration, from multi-pixel measurements by consolidating the strongest variance component into a principal breathing waveform.

Through this joint iterative localization and interference mitigation, MURAL-Fi robustly uncovers each user's true breathing rhythm despite dynamic environments and multi-user interference. By the time the iterative process converges, MURAL-Fi has isolated stable, interference-free chest cavity regions and accurate respiration waveforms suitable for subsequent analyses. In this section, both the input and output are time-series data. The input is obtained from Algorithm 1 using the continuous 2D AoA spectrums, while the output is interpolated to 50 Hz to ensure temporal alignment and consistency with the input.

8 AUTHENTICATION MODEL

8.1 Multi-level Feature Extraction

The input instances used in the SNN contain only respiration-related features, which are waveforms derived from the user's breathing motion through our system. No other spatial spectrum information is included. Our authentication system leverages three complementary techniques to extract distinctive features from respiratory motion signals: waveform analysis, derivative analysis, and DWT Analysis.

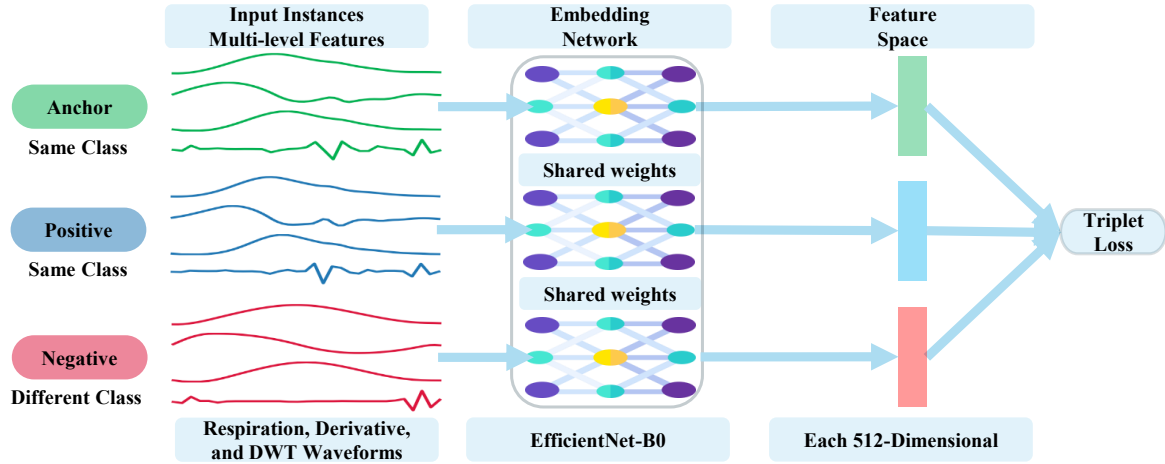


Fig. 6. Architecture of the MURAL-Fi triplet loss Siamese Neural Network.

Waveform Analysis: Research has shown that a single breathing cycle provides insufficient differentiation for authentication [25, 44]. As shown in Fig. 6, we can observe a distinct respiration pattern extracted by our system. Therefore, we segment the constructed signals into multiple periodic breathing cycles to capture more distinctive characteristics. To maintain consistency across varying cycle lengths, we interpolate all segments to match the length of the longest breathing cycle. This interpolation preserves the detailed morphology of respiratory patterns while enabling direct comparison across different segments.

Derivative Analysis: To capture the dynamic characteristics of breathing patterns, we compute derivatives of the aligned respiratory waveforms. This analysis reveals crucial temporal features such as instantaneous changes in breathing velocity and acceleration. The derivative analysis is particularly effective at highlighting subtle inter-individual variations in breathing patterns while suppressing common noise components, thereby enriching the feature set for authentication.

DWT Analysis: We employ 1D DWT to extract features that are sensitive to both temporal and frequency characteristics of the breathing signals. The DWT decomposes the signal into approximation coefficients (low-frequency components) and detail coefficients (high-frequency components). Through iterative decomposition up to three levels, we obtain a multi-resolution representation that captures respiratory features at different frequency bands. This hierarchical decomposition effectively balances noise reduction with feature preservation, focusing on frequency ranges that contain stable and discriminative respiratory characteristics.

We combine the analysis results into a composite image following the order of the user-specific respiration waveform, derivative analysis, and DWT representations, including both approximation (low-frequency) and detail (high-frequency) components. These waveforms, arranged sequentially, serve as the input to an SNN for further processing.

8.2 Siamese Neural Network

For multi-user authentication, we implement an SNN trained with triplet loss [34]. The SNN is specifically designed to learn and compare similarities [11] between pairs of respiratory features, making it ideal for authentication tasks[3]. The network training process utilizes triplets of instances: an anchor (x_{a_i}), a positive sample (x_{p_i}), and a negative sample (x_{n_i}). The positive sample comes from the same individual as the anchor, while the negative sample comes from a different individual. These instances are processed through the same feedforward network

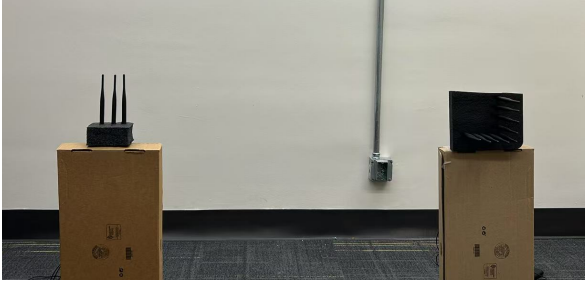


Fig. 7. Experiment WiFi devices.

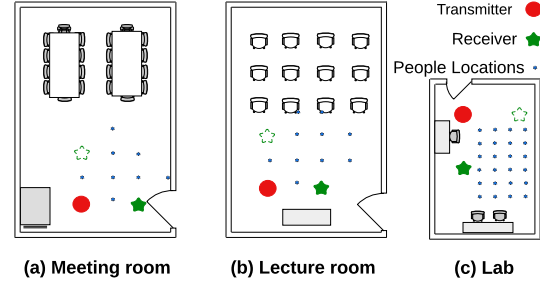


Fig. 8. Experiment environments and setups.

F to generate their embedded representations. The training objective is to ensure the anchor's embedding is closer to the positive sample's embedding than to the negative sample's. This is achieved by minimizing the triplet loss:

$$L = \sum_{i=1}^N \left[|F(x_{a_i}) - F(x_{p_i})|_2^2 - |F(x_{a_i}) - F(x_{n_i})|_2^2 + \alpha \right]_+ \quad (9)$$

where α represents the margin enforced between positive and negative pairs [34]. This margin ensures sufficient separation between embeddings of different individuals while maintaining compact clusters for the same individual.

During the enrollment phase, the trained network processes the multi-level respiratory features of legitimate users to create their embedding profiles. These profiles are stored as reference vectors for subsequent authentication. The authentication process then operates continuously by analyzing incoming respiratory signals and comparing their embeddings against the stored profiles using Euclidean distance as the similarity metric.

For effective multi-user authentication, the network is trained to generate embeddings that satisfy two key criteria: (1) respiratory signals from the same individual should produce closely clustered embedding vectors, and (2) signals from different individuals should result in widely separated embeddings. When authenticating a user, the system computes the distances between their current embedding and all stored profile embeddings. The identity associated with the smallest distance is considered the closest match, enabling accurate user identification. In this section, the input is time-series data from interference suppression. The output is the embedding features. Since breathing cycle durations naturally vary among individuals, we interpolate each segment to align with the longest breathing cycle in the dataset to ensure consistent input length. Finally, multi-level features are visualized and merged into a $128 \times 64 \times 3$ image, which serves as the input instance for the Siamese neural network.

9 PERFORMANCE EVALUATION

9.1 Experimental Setups

9.1.1 Devices Setups. In our experiments, we implement MURAL-Fi using commodity WiFi devices. Specifically, we use Dell LATITUDE laptops equipped with Intel 5300 network interface cards (NICs) on both the transmitter and receiver sides. The transmitter is equipped with a linear array of three antennas. The receiver is equipped with nine antennas forming an L-shaped antenna array, as shown in Fig. 7. This L-shaped antenna array consists of two uniform linear subarrays positioned orthogonally. Each subarray includes two NICs interconnected via a signal splitter that shares a common antenna. We use the common antenna to synchronize the phase shift among NICs. This configuration emulates potential antenna setups of next-generation WiFi devices, for instance, WiFi 7

supports up to 16 antennas. Such an L-shaped layout can be realized on commercial devices by arranging multiple antennas along different edges of the device enclosure. The antenna geometry is flexible since our MUSIC-based algorithm does not assume any array geometry [33]. The receivers operate in monitor mode to capture packets from the transmitter. Both the transmitting and receiving antennas are uniformly spaced at half a wavelength (2.8cm). The WiFi channel is set to a frequency of 5.24 GHz with a bandwidth of 40 MHz, and the transmitter sends 1000 packets per second. We employed the Linux 802.11 CSI tool [14] to acquire CSI measurements from 30 OFDM subcarriers. We use this tool to validate our algorithms under real-world wireless conditions. Our system is not limited to this tool and can be extended to any CSI-enabled WiFi platforms, aligning with the sensing objectives of the emerging IEEE 802.11bf standard.

9.1.2 Data Collection. To evaluate our system, we recruited 24 participants (14 male and 10 female) aged 21–54 years, with heights ranging from 1.55 m to 1.92 m and weights from 48 kg to 90 kg, representing diverse body builds. All participants were monitored in their natural resting state, with breathing rates varying between 10 to 30 breaths per minute. Experiments were conducted in three different environments: a meeting room, a lecture room, and a lab room, as shown in Fig. 8. The meeting room includes two long tables, multiple chairs, and a projector. The room size is 6m × 9.5m. The lecture room contains multiple pairs of desks and chairs, a long lectern, and measures 6m × 9.5m. The lab room size is 4m × 6m. The lab room contains 4 desktop computers, desks, and chairs. Across all three locations, the desks and tables were wooden, while the chairs were made of metal and plastic. As shown in Fig. 8, the distance between the transmitter and receiver is 2 m. The distances between the adjacent dots are 1 m in environments meeting room and lecture room, and 0.5 m in the Lab. Each room has a different layout, size, and furniture such as chairs, tables, and devices. Different environments are used to evaluate respiration features and system performance under varied real-world environmental factors. For the WiFi CSI dataset collection, each sample corresponds to a 30-second CSI segment. We record 10 to 60 samples per user in different evaluation scenarios. Over six months, we collected around 55 million WiFi CSI packets to evaluate our system. The data collection was approved by the IRB of the authors' institution.

9.1.3 Baselines. **Cuvrb** [25] proposes a user verification system that extracts morphological features and fuzzy wavelet packet-based features from CSI amplitude. To extract these features, they remove noise with an EMD-based filter and select subcarriers based on periodicity activities.

MultiSense [48] proposes a multiple-person respiration monitoring system. They first divide the raw CSI between two antennas to remove time-varying phase offset and select subcarriers. Then they apply for a RobustICA to separate multi-person signals. Finally, they combine segmentations by an optimal matching algorithm.

MUSE-Fi [18] introduces a WiFi system for multi-person sensing. By exploiting the near-field channel variation from subjects close to their devices, MUSE-Fi achieves physical separability among users because the user who is close to the receiver will dominate the signal variation of the receiver. They evaluate multiple applications including respiration monitoring however, this system requires multiple receivers.

9.1.4 Model Settings. The model input is from multi-level features combined as 2d matrix image-like data. The embedding model is EfficientNet-B0 [37]. The final classification layer was replaced with a linear layer producing embeddings of size 512. We set a batch size of 32. We utilize triplet margin loss and Adam optimizer with a learning rate of 0.01. To compare the distance between embeddings, we utilize Euclidean distance. We implement our deep learning model using PyTorch and train it on an NVIDIA RTX 3090 GPU. To train the Siamese Neural Network (SNN), we selected at least 12 users, each contributing over 50 WiFi signal samples. The dataset was divided into 70% for training, 20% for validation, and 10% for testing, ensuring no overlap among the three subsets. It is important to note that our system does not require retraining for each evaluation scenario. Once the model is trained, it can directly evaluate authentication performance under different conditions without additional model retraining.

9.1.5 User Authentication. For authentication, the system begins with an enrollment process and operates with the authentication procedure described in Section 2.2. We examine the embedded features between monitoring respiration signals and the user's profile. The user profile contains 12 to 18 users, randomly selected from participants. Intuitively, if the monitoring respiration signals belong to the legitimate user, the Euclidean distance between the monitoring user and the legitimate user should be small. Conversely, if the monitoring respiration signals originate from an unauthorized individual, the distance should be significantly larger. Based on empirical analysis in Section 9.5, to balance authentication accuracy and latency, we adopt three respiration cycles as input to the authentication process, providing a practical trade-off between system performance and responsiveness. To prove the authentication capability of our system, the evaluation involves a total of 24 participants, where each participant acts as the registered owner, with the remaining participants serving as attackers [24]. A threshold is employed to distinguish between enrolled users and non-enrolled users [25].

9.1.6 Evaluation Metrics. Authentication accuracy is measured using the F1 score. This metric provides a more reliable accuracy evaluation, especially when the true class distribution is imbalanced. F1 score is defined as follows:

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * TP}{2 * TP + FP + FN}, \quad (10)$$

where the True Positive (TP) occurs when the system correctly identifies an authorized user as legitimate. False Positive (FP) occurs when the system incorrectly identifies an unauthorized user as legitimate, granting attackers access. False Negative (FN) occurs when the system incorrectly identifies an authorized user as unauthorized, denying owner access. The Receiver Operating Characteristic (ROC) curve describes the trade-off between the true positive rate (TPR) and the false positive rate (FPR) across varying thresholds.

We evaluate the similarity between the extracted user-specific respiration waveform and reference waveform from the contact device [29] using Cosine Similarity, which quantifies how closely two waveforms align in direction regardless of their magnitude. The cosine similarity between the extracted signal E and the reference signal R from the contact device is defined as

$$\text{Similarity}(E, R) = \frac{E \cdot R}{\|E\| \times \|R\|}. \quad (11)$$

9.2 Multi-user Authentication

In this section, we evaluate our system under varying numbers of users (from 1 to 5) to assess its performance in multi-user scenarios. The results are compared against both state-of-the-art respiration-based authentication systems [25] and multi-user respiration sensing approaches [18, 48]. All users are static and breathing normally. Therefore, all of them are target people to be authenticated. Note that only one user is enrolled and registered as a legitimate user. The other are enrolled and registered as illegitimate users. We assume that only one user is legitimate at a time, simulating a realistic setting where multiple users are present in the same environment. The personal device is designed to grant access exclusively to the legitimate user, even in the presence of other illegitimate users. At each moment, a maximum of five users were authenticated. We collect 15000 seconds of CSI data in this scenario. We compare our approach with all baseline systems under the same experimental setup. The results in Fig. 9 show that Cuvrb [25] achieves authentication accuracies of 84.5%, 40.3%, 24.1%, 17.1%, 10.2% for 1 to 5 people. While effective for single-user authentication, the system's performance degrades significantly in the presence of additional users, due to its exclusive reliance on amplitude information. This system lacks the ability to separate multi-user respiration signals. Multisense achieves authentication accuracies of 93.7%, 45.3%, 29.0%, 21.1%, and 16.2% for 1 to 5 people. While it performs reasonably well for single-person scenarios, its performance degrades significantly as the number of people increases. Although Multisense is capable of separating multi-user

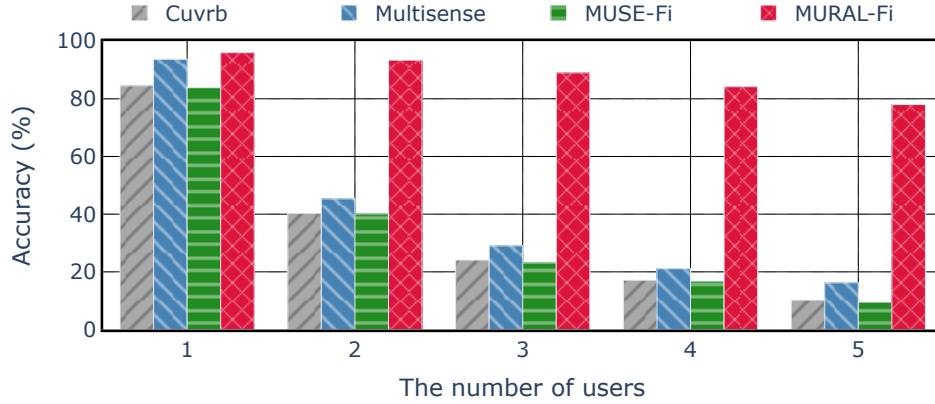


Fig. 9. Comparison and analysis on MURAL-Fi and state-of-the-art systems under multi-user scenarios.

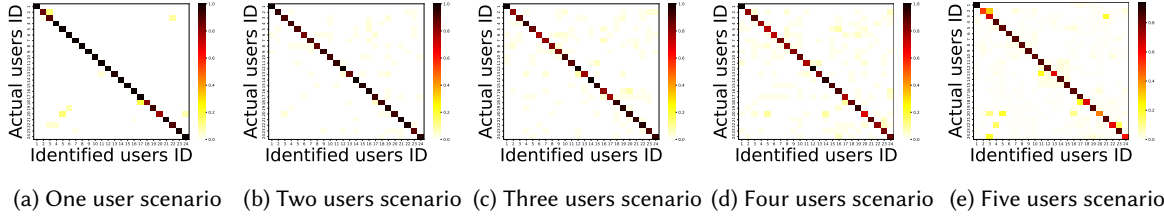


Fig. 10. Confusion matrices for multi-user scenarios ranging from a single user to five users, illustrating MURAL-Fi classification performance across different numbers of users.

respiration signals and extracting refined phase information from CSI, it struggles to correctly associate each signal with its corresponding individual. This limitation leads to increased false positives and false negatives as the number of users grows. The MUSE-Fi achieves authentication accuracies of 83.9%, 40.1%, 23.1%, 16.8%, and 9.5% for 1 to 5 people. Although the system can physically separate multiple users, MUSE-Fi relies on multiple User Equipments (UE). In our experimental setup, we use only a single receiver, which limits MUSE-Fi's ability to effectively separate multiple users. Multisense performs better than Cuvrb and MUSE-Fi in single users because Multisense cancels out the time-varying phase offset leveraging antenna pairs. After this cancellation, they refine the phase information from CSI. Multisense also performs better than Cuvrb and MUSE-Fi in multi-users. Because they use an ICA-based algorithm to separate multiple people's respiration motion signals.

Our system achieves authentication accuracies of 96.0%, 93.4%, 89.0%, 84.2%, and 78.0% for 1 to 5 people, respectively. Our system outperforms all baseline systems in multi-person scenarios by identifying the chest cavity region of each user and effectively mitigating interference caused by other users' respiration. With a fine-grained waveform feature extraction of each user, our system achieves an identity-aware authentication. Our system maintains strong performance with up to three individuals. Multi-user sensing is challenging due to signal overlap and mutual interference in shared environments. Nevertheless, our system continues to outperform other methods even as the number of users increases to four and five. We also demonstrate that increasing the number of respiration cycles can enhance performance in multi-user scenarios in Section 9.5. In fig. 10, we also present

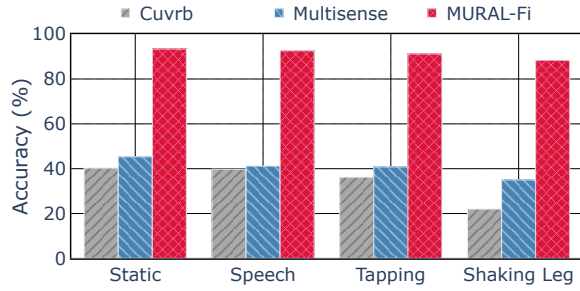


Fig. 11. Comparison and analysis of MURAL-Fi and the baselines under self-involuntary motion interference scenarios.

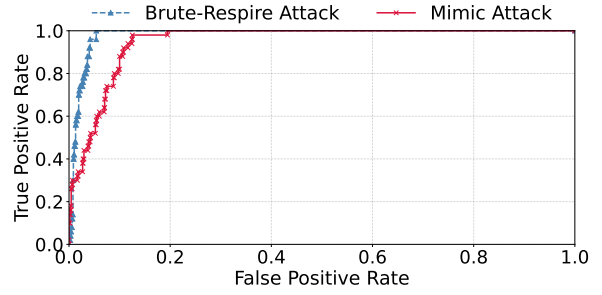


Fig. 12. ROC curves illustrating MURAL-Fi performance under brute-respire and mimic attack scenarios.

color-scale maps of the confusion matrices for user identification, illustrating the distribution of identification errors.

9.3 Involuntary Motion Tolerance

To evaluate the impact of self-involuntary motion interference, we conducted experiments in which two users performed routine motions such as static, speech, tapping, and leg shaking. Specifically, tapping denotes finger or hand tapping on the keyboard, while leg shaking refers to rhythmic oscillation of the leg. These activities were selected to represent natural body motions that may occur daily. One is a legitimate user, the other is an illegitimate user. We collect 6000 seconds of CSI data in this scenario. These users maintained normal breathing for authentication. We stored profiles for 12 enrolled users, and to evaluate authentication integrity, the user to be authenticated included both enrolled and non-enrolled users. The distance between two users is from 50 cm to 2 m. The results of our system are shown in Fig. 11. Cuvrb achieves authentication accuracies of 40.3%, 39.8%, 36.2%, and 22% under static, speech, tapping, and leg shaking scenarios, respectively. The performance of Cuvrb is the worst in interference scenarios since the frequencies of self-involuntary motion can overwhelm the frequencies of respiratory signals. Multisense achieves authentication accuracies of 45.3%, 41.2%, 40.8%, and 35% under static, speech, tapping, and leg shaking scenarios, respectively. The performance of Multisense is better than Cuvrb as it can separate respiration signals. However, it is still sensitive to non-linear interferences. Our system demonstrates superior performance compared to the other two systems. Specifically, our system achieves the highest accuracies of 93.4%, 92.6%, 91.2%, and 88.4% under static, speech, tapping, and leg shaking scenarios, respectively. The high accuracy of our system is primarily due to its ability to leverage spatial information extracted from 2D AoA spectrums to accurately identify each user's chest cavity region. By focusing on signals from these regions, the system effectively mitigates self-induced interference caused by movements of other body parts, such as the legs and arms, as well as interference from other users' motions. Furthermore, our system incorporates an adaptive interference mitigation technique, ensuring robust separation of respiratory signals from noise and distractions. As a result, the result demonstrates significantly greater resilience to interference compared to Cuvrb and Multisense, achieving superior performance in self-involuntary motion interference scenarios.

9.4 System Performance under Attacks

To evaluate the robustness of our respiration-based WiFi authentication system against adversarial attempts, we conduct experiments under two attack scenarios: brute-respire attack and mimic attack. In the brute-respire attack,

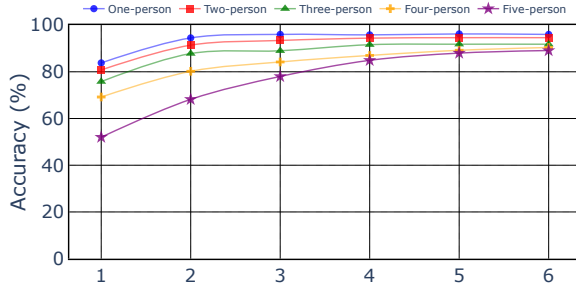


Fig. 13. Analysis of MURAL-Fi performance across different breathing cycles.

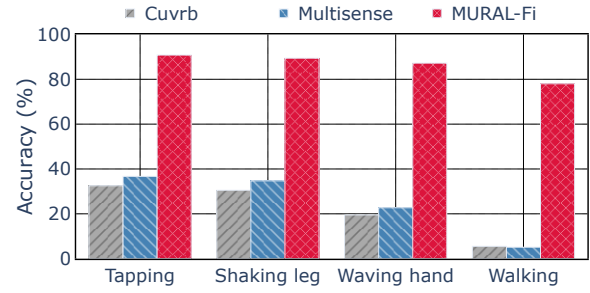


Fig. 14. Comparison and analysis on MURAL-Fi and the baselines under multiple users and interference scenarios.

adversaries attempt authentication by randomly replaying general respiration patterns without any knowledge of the legitimate users' breathing characteristics. In contrast, the mimic attack assumes a stronger adversary who actively observes the target user's respiration behavior and attempts to imitate it to deceive the system.

We evaluate system performance under these two types of attacks and plot the corresponding Receiver Operating Characteristic (ROC) curves, as shown in Fig. 12. Specifically, in the brute-respire attack, the attack detection rate is over 96% and the authentication system maintains a high TPR and low FPR, indicating that random breathing patterns are unlikely to pass as legitimate users. In the mimic attack, the attack detection rate is over 93% and our system still achieves a high TPR and low FPR. The slight decrease in detection rate observed under mimic attacks suggests that imitating a victim's breathing pattern may marginally aid adversaries. However, precisely replicating another individual's respiration dynamics remains inherently challenging due to natural variations in breathing, such as unique turbulent flow patterns in exhaled breath [12]. Additionally, increasing the number of respiration inputs further enhances our system's detection accuracy, as demonstrated in Section 9.5. Overall, these results confirm that the system maintains high robustness against both brute-respire attack and mimic attack, preserving secure authentication performance under attacks.

9.5 Impact of Breathing Cycles

In this section, we evaluate our system across varying numbers of users (from one to five) collecting multiple breathing cycles. As shown in Fig. 13, the system achieves its highest accuracy of 96.2%, 94.4%, 92.0%, 91.4%, and 90.2% for multiple users from 1 to 5 users, respectively. Authentication accuracy improves with collecting more respiration cycles across all user group sizes. While accuracy declines with more users due to increased signal overlap and interference, the system still maintains competitive performance, achieving over 90% accuracy even in the five-user case. This demonstrates the scalability and reliability of our method in challenging multi-user settings.

9.6 Environment Interference Robustness

To evaluate the impact of interference, we conducted experiments in which one person performed daily activities such as tapping, leg shaking, hands waving, and walking, while the other remains at normal breathing to be authenticated. The results of our system are shown in Fig. 14. Only the normally breathing user was to be authenticated. Our system processes the WiFi signals of a normally breathing user and extracts embedding features. These embeddings are then compared against the stored enrollment profile. If the similarity score exceeds a predefined threshold, the system identifies the individual as an enrolled user. Otherwise, if the similarity score falls below the threshold, the system classifies the individual as a non-enrolled user. The second

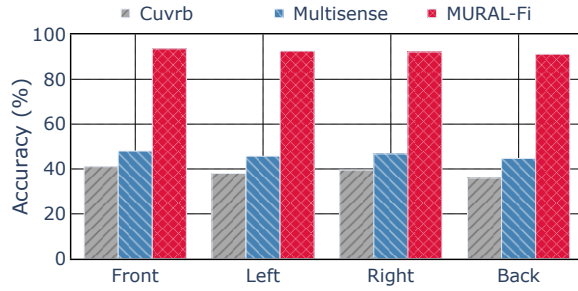


Fig. 15. Comparison and analysis on MURAL-Fi and the baselines under different orientations scenarios.

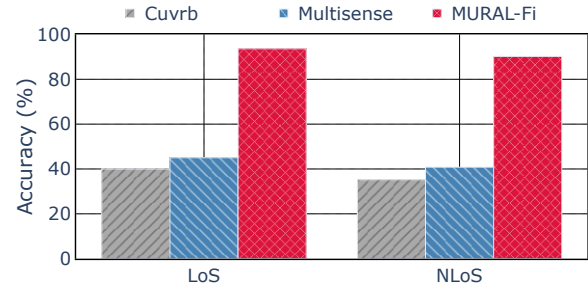


Fig. 16. Comparison and analysis on MURAL-Fi and the baselines under non-line-of-sight scenarios.

participant was included solely to generate environmental interference. We collect 6000 seconds of CSI data in this scenario. Cuvrb achieves authentication accuracies of 32.5%, 30.5%, 19.5%, 5.5% under typing, shaking legs, waving hands, and walking scenarios, respectively. Cuvrb performance drops in interference scenarios, as the frequency components of interference often dominate and obscure those of the respiration signals. Multisense achieves authentication accuracies of 36.7%, 34.7%, 22.7%, and 5.1% tapping, leg shaking, hands waving, and walking scenarios, respectively. Multisense outperforms Cuvrb by separating respiration signals from multiple users through independent component analysis. However, it remains sensitive to non-linear interference, which can distort the separation process. In contrast, our system achieves superior performance under both linear and non-linear interference conditions, demonstrating greater robustness and reliability. Specifically, our system achieves the highest accuracies of 90.8%, 89.4%, 87.2%, and 78% under tapping, leg shaking, hands waving, and walking scenarios, respectively. This improvement is attributed to our system's use of spatial information derived from 2D AoA spectrums, which enables effective separation of respiration signals from other individuals' motions. Furthermore, we incorporate an interference mitigation technique that enhances signal quality. As a result, our system exhibits greater robustness against interference compared to both Cuvrb and Multisense.

9.7 Impact of User Orientation

To evaluate the impact of different orientations on system performance, we conduct experiments to evaluate how variations in body orientation influence the amplitude of displacement caused by breathing. In the experiment, we ask for two individuals to sit in four orientations relative to the transmitter: facing in front of the transmitter, with their backs to the transmitter, and oriented toward the left and right sides of the transmitter. One is a legitimate user, the other is an illegitimate user. The results, presented in Fig. 15, illustrate that Cuvrb achieves the authentication accuracy of 41% in front of the transmitter, 38% for the left side, 39.5% for the right side, and 36% back to the transmitter. Multisense achieves the authentication accuracy of 48% in front of the transmitter, 45.7% for the left side, 46.7% for the right side, and 44.7% back to the transmitter. Our system achieves 93.6% in front of the transmitter, 92.6% for the left side, 92.2% for the right side, and 91.2% back to the transmitter. This highlights the robustness of our system in handling variations in multiple users' body orientations. The superior performance of our system can be attributed to its ability to extract respiratory features accurately, regardless of the direction of signal propagation. The minimal degradation in accuracy, even when users are not directly facing the transmitter, underscores the system's effectiveness in mitigating signal distortion and ensuring reliable performance in real-world scenarios. Our system has a better performance than two baselines mainly resulting in two users in the environment.

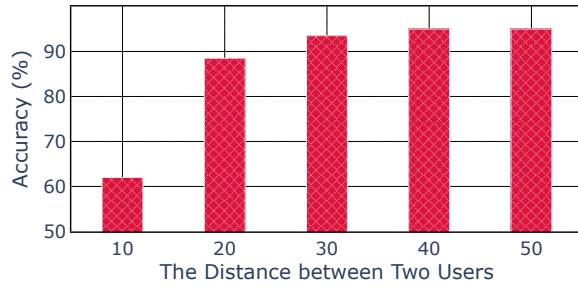


Fig. 17. System performance over different distances between two users.

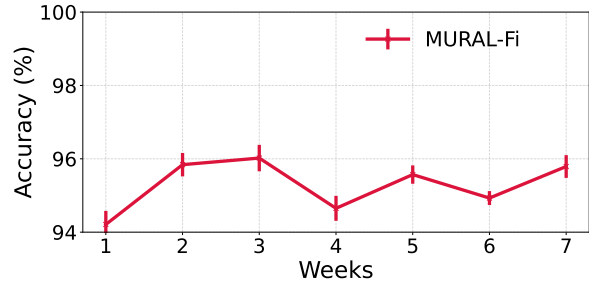


Fig. 18. System performance over a 7-week longitudinal study.

9.8 Non-Line-of-Sight Authentication

In this section, we evaluate the system's performance in Non-Line-of-Sight (NLoS) scenarios by using a wooden panel as an obstacle positioned between the WiFi devices and two users. One is a legitimate user, the other is an illegitimate user. We collect 2400 seconds of CSI data in this scenario. The results are illustrated in Fig. 16. Specifically, under NLoS scenarios, Cuvrb achieves authentication accuracies of 35.2%. Multisense achieves accuracies of 40.8% under NLoS scenarios for authentication, respectively. In contrast, our system demonstrates superior performance, achieving an authentication accuracy of 90.2%. Across all three systems, performance under NLoS scenarios is lower compared to LoS scenarios due to signal attenuation caused by obstacles, which reduces the Signal-to-Noise Ratio (SNR) of the received signals. Nevertheless, our system outperforms the other baselines by effectively separating the respiratory signals of multiple individuals and mitigating interference, even in challenging NLoS conditions. These results show that our system works well for NLoS scenarios, where the traditional computer vision-based system won't work.

9.9 Impact of Distance between Users

To evaluate the impact of distance between two users, we conducted experiments to evaluate our system's performance when two users are positioned at close angular separations with different inter-user distances. One as a legitimate user, the other as an illegitimate user. We collect 7500s of CSI data in this scenario. Specifically, the illegitimate user was positioned 1 m away from the WiFi receiver, perpendicular to the LoS, while the legitimate user was placed parallel to the LoS at distances of 10 cm, 20 cm, 30 cm, 40 cm, and 50 cm from the illegitimate user. We collect 7500 seconds of CSI data in this scenario. As shown in Fig. 17, when the distance between the two users was 10 cm, 20 cm, 30 cm, 40 cm, and 50 cm, the legitimate user's authentication accuracies were 62.0%, 88.5%, 93.5%, 95.0%, and 95.0%, respectively. These results indicate that as the inter-user distance increases, the system's ability to distinguish individual respiration patterns improves. When two users are within 10 cm of each other, their reflected signals overlap, causing interference that hampers the accurate extraction of user-specific spatial and temporal features. As the separation widens, the independent propagation paths corresponding to each user become more distinguishable in the multiple-dimension domain, enabling the system to isolate respiration-induced motion more effectively. This demonstrates that our joint estimation of spatial parameters enhances robustness against mutual interference, even for 30cm distance separations, and confirms the system's capability to maintain reliable performance in multi-person environments.

Table 2. Overall system performance comparison without and with data calibration.

	Cosine Similarity	Accuracy
System without data calibration	0.82	91.3%
System with data calibration	0.95	96.2%

9.10 Impact of Longitudinal Study

In this section, we evaluate the system's long-term stability by conducting authentication tests over a continuous seven-week period. We collect 2100 seconds of CSI data in this scenario. As shown in Fig. 18, our system maintains consistent performance throughout the study, achieving an average authentication accuracy of around 95.5%, with a standard deviation of less than 0.4%, indicating only minor fluctuations across weeks. The results indicate that, once enrolled one time, a user does not need to re-enroll even after several weeks, confirming the persistence of the extracted respiration-related features. Overall, these findings confirm that our system provides stable and reliable authentication over extended durations, underscoring its practicality for long-term deployment.

10 ABLATION STUDY

To better understand our system workflow and evaluate the effectiveness of each stage, we divide the framework of our system into a series of processing stages, including data calibration, multi-dimensional information combination, and interference suppression, and utilize the ablation study to evaluate each stage.

First, to evaluate the effectiveness of the data calibration process, we compare the overall system performance with and without data calibration. Table 2 shows our system without data calibration achieves similarity scores of 0.82 and authentication accuracies of 91.3%. System with data calibration achieves similarity scores of 0.95 and authentication accuracies of 96.2%. The results demonstrate that our calibration process removes random phase offsets, which benefits subsequent AoA and ToF estimation and further improves the overall performance.

Second, to verify the effectiveness of multi-dimensional information fusion, we compare the results of our system with four baselines to evaluate how additional spatial dimensions enhance system performance. Specifically, Baseline A only estimates AoA, Baseline B estimates AoA + ToF, Baseline C estimates 2D AoA (azimuth and elevation) + ToF, and Baseline D estimates 2D AoA + ToF + AoD. Our system further leverages 2D AoA + ToF + AoD + Doppler to achieve higher signal resolvability. Table 3 presents the comparison of the overall system performance for two to five subjects between MURAL-Fi and the four baseline methods. Baseline A achieves a cosine similarity score of 0.76 and an authentication accuracy of 82.6%, as it only utilizes one-dimensional AoA information, which is insufficient for distinguishing multiple users as lower angle resolution. Baseline B improves the similarity score to 0.79 and accuracy to 88.5% by incorporating ToF information, which provides additional spatial resolution. Baseline C achieves a similarity score of 0.93 and an accuracy of 92.0%, benefiting from extending AoA estimation from 1D to 2D (azimuth and elevation) to better localize each subject's chest region. Baseline D maintains a similarity score of 0.93 and slightly improves accuracy to 93.2% by introducing AoD, enabling differentiation among users with AP view. Finally, MURAL-Fi achieves the highest performance, with a similarity score of 0.95 and an authentication accuracy of 96.2%, by jointly incorporating Doppler along with AoA, ToF, and AoD, thereby enhancing spatial resolution, signal resolvability, and user separability.

Third, to evaluate the effectiveness of interference suppression, we compare the overall system performance with and without interference suppression. Table 4 shows that the system without interference suppression achieves a cosine similarity of 0.74 and an authentication accuracy of 81.4%, while the system with interference suppression achieves a cosine similarity of 0.95 and an authentication accuracy of 96.2%. The results demonstrate

Table 3. Overall system performance comparison of multi-dimensional spatial information.

	Dimensions	Cosine Similarity	Accuracy
Baseline A	AoA	0.76	82.6%
Baseline B	AoA + ToF	0.79	88.5%
Baseline C	2D AoA + ToF	0.93	92.0%
Baseline D	2D AoA + ToF + AoD	0.93	93.2%
MURAL-Fi	2D AoA + ToF + AoD + Doppler	0.95	96.2%

Table 4. Overall system performance comparison of without and with interference suppression.

	Cosine Similarity	Accuracy
System without interference suppression	0.74	81.4%
System with interference suppression	0.95	96.2%

that our interference suppression process effectively mitigates interference, including environmental noise, cross-user interference, and involuntary motions, significantly enhancing respiration reconstruction and authentication accuracy.

11 RELATED WORK

Authentication technologies utilize diverse sensing modalities for reliable user authentication. Biometric-based methods can be grouped into contact-sensor-based, RF Sensing-based, and other modalities-based approaches.

Contact-sensor-based Approaches. Contact-sensor-based authentication methods capture biometric data through direct physical interaction with sensors. Sim et al.[35] proposed a multimodal biometric verification system using face and fingerprint data, requiring users to interact continuously with the device for ongoing authentication. Wang et al.[41] utilized a smartphone's built-in accelerometer to capture heartbeat signals for authentication. Chauhan et al. [7] explored breathing acoustics-based authentication on smartphones, smart-watches, and Raspberry Pi devices, showing that compact LSTM models achieve high accuracy and low latency even under resource constraints. However, the need for users to hold the device motionless limits the practicality and user-friendliness of these systems for continuous use.

RF Sensing-based Approaches. In recent years, radio frequency (RF) sensing techniques have been widely used for applications such as authentication [26, 52]. RF-based methods leverage RF signals to capture user-specific physiological or behavioral traits without requiring physical contact. For example, Lin et al. [24] introduced CardiacScan, a continuous authentication system using CW Doppler radar to extract heartbeat waveforms. While effective in capturing fine-grained signals, it lacks tolerance to involuntary motion and is limited to single-user scenarios. Hu et al. [17] introduce a low-cost radio-frequency identification system that measures breathing pattern at the chest and abdomen and performs continuous user verification. Wang et al.[44] and Kong et al.[20] used mmWave radar and CSI, respectively, to achieve high authentication accuracy based on physiological features. However, both systems require dedicated hardware or specific user actions, limiting their practicality in multi-user or continuous scenarios. Liu et al.[25] proposed a WiFi-based respiratory authentication system but struggled to separate overlapping signals in multi-user environments, reducing reliability. Similarly, Zeng et al.[48] developed a multi-user respiration monitoring system using WiFi. Unfortunately, these systems lack

identity awareness and therefore cannot distinguish use physically, making them unsuitable for authentication purposes.

Other Modalities-based Approaches. The increasing availability of vision datasets has driven the development of computer vision-based authentication systems. Chen et al. [9] proposed a face authentication system using mobile phone cameras, incorporating photoplethysmograms for liveness detection to enhance security. Zulfiqar et al. [53] introduced a facial biometric authentication system based on a Convolutional Neural Network (CNN), achieving robustness against variations in scale, translation, rotation, and lighting. Fong et al. [13] extracted unique behavioral characteristics from gesture images in sign language, using machine-learning models for authentication. However, such systems are ineffective in non-line-of-sight (NLoS) conditions or environments with poor lighting. Acoustic sensing enables user authentication by capturing physiological features through sound signal reflections. Lu et al. [28] leverages vocal tract characteristics for passphrase-independent authentication on smartphones, offering robustness against replay attacks. Wu et al. [45] utilizes acoustic echoes to authenticate users based on hand geometry and movement, achieving high accuracy and resistance to presentation attacks. Chen et al. [10] combines voice and chest motion biometrics via acoustic sensing to resist replay attacks on smart devices, achieving high authentication accuracy with few samples using meta-learning. Wang et al. [42] introduced MultiResp, a robust acoustic sensing framework that identifies multiple paths by frequency and phase. Wan et al. [40] proposed RespTracker, which utilizes inaudible sound signals modulated by the Zadoff-Chu (ZC) sequence to separate acoustic reflections from different users. By leveraging path distance information, RespTracker effectively distinguishes and reconstructs multiple users' respiration signals for monitoring. Above mentioned methods cannot be directly applied to WiFi systems, which rely on OFDM-based signal. In contrast, WiFi provides an existing infrastructure platform, unlike other sensing modalities that require dedicated hardware and complex setup in typical indoor environments. Our work is the first WiFi-based authentication system that physically identifies legitimate users based on their respiration patterns.

12 DISCUSSION

In this section, we present and discuss several limitations and potential future work of this work.

Crowded Scenario. Our current system can accurately reconstruct respiration patterns and authentication for multiple individuals. However, this capability does not imply that the number of detectable subjects is unlimited. In densely populated indoor environments, the system's accuracy decreases when individuals are positioned very close to one another and completely overlap or occlude from the perspective of the WiFi devices. In such configurations, reflected signal paths overlap with each other, reducing the SNR of the received WiFi signals and hindering reliable separation and tracking of each subject's respiration. Consequently, precise authentication for individual users becomes challenging when two users fully overlap or occlude. Nevertheless, during typical indoor authentication, from a security perspective, when our system cannot confidently identify a legitimate user in a crowded area, our system can physically separate users by region and reject all individuals within the crowded area. As a result, the influence of these overlaps on the overall authentication performance remains limited. Looking ahead, as modern WiFi standards continue to evolve with higher antenna counts and improved spatial resolution, incorporating additional antennas or multiple WiFi transceivers could further enhance signal separability and mitigate the limitations currently observed in crowded scenarios.

Physiological Changes. This study focuses on authentication and does not consider physiological variations (e.g., stress or exercise). All participants breathed naturally without intervention to ensure safety. Future work could examine how physiological changes affect respiration-based authentication.

Real-time Deployment. Our current deployment is unable to achieve real-time processing on consumer-grade hardware. However, investing time to ensure that the system's authentication process is trustworthy against potential attacks is essential for maintaining overall system security and reliability. Our system's computational

cost primarily arises from three components: the multidimensional MUSIC-based algorithm, the EMD-based interference suppression, and the SNN. Among them, the multidimensional MUSIC-based algorithm and EMD-based interference suppression dominate the computation complexity. However, the MUSIC-based and EMD-based processes can be run in parallel, and the MUSIC-based algorithm can be executed in reduced dimensions to improve efficiency. The SNN introduces negligible computational overhead after training.

13 CONCLUSION

In this paper, we presented MURAL-Fi, the first system to achieve multi-user respiration-based authentication using WiFi, explicitly addressing three key requirements: multi-user support, identity-aware capability, and involuntary motion tolerance. Central to MURAL-Fi is a novel statistical approach that identifies a region of strong energy points in the WiFi spectrum, rather than tracking only the single highest-energy peak. This regional perspective, coupled with our iterative region refinement method, robustly captures the user's chest movements while suppressing environmental noise and involuntary motion artifacts. Our triplet loss-based Siamese Neural Network further ensures accurate and efficient authentication. Extensive experiments with 24 participants across multiple indoor environments demonstrate that MURAL-Fi maintains high accuracy and robustness under challenging real-world conditions, including variable distances, user orientations, and multiple users. Overall, MURAL-Fi pushes the boundaries of non-intrusive continuous authentication by leveraging ubiquitous WiFi infrastructure, offering a practical and scalable solution for next-generation security systems.

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